

WOLFGANG WILLIAM SCHMALTZ

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EDUCATION

University of California, Berkeley

Aug. 2012 – May 2018

Doctor of Philosophy in Mathematics

University of Chicago

Sep. 2007 – Jun. 2011

Bachelor of Science in Mathematics

RESEARCH POSITIONS

Polyfold Theorist

Mar. 2020 – Dec. 2024

Faculty of Mathematics, Ruhr-Universität Bochum

Bochum, Germany

Polyfold Theorist

Jan. 2018 – Feb. 2020

Mathematics Institute, Justus-Liebig University

Gießen, Germany

PAPERS

Pseudocycle Gromov–Witten invariants are a strict subset of polyfold Gromov–Witten invariants

Aug. 2023

arXiv:2308.14204

W. Schmaltz

Non-fillability of overtwisted contact manifolds via polyfolds

Nov. 2020

arXiv:2011.02249

W. Schmaltz, S. Suhr, K. Zehmisch

The Gromov–Witten axioms for symplectic manifolds via polyfold theory

Dec. 2019

arXiv:1912.13374

W. Schmaltz

Naturality of polyfold invariants and pulling back abstract perturbations

Dec. 2019

arXiv:1912.13370

W. Schmaltz

The Steenrod problem for orbifolds and polyfold invariants as intersection numbers

Apr. 2019

arXiv:1904.02186

W. Schmaltz

SELECTED INVITED TALKS

Orbifold Intersection Theory and Polyfold Gromov–Witten Invariants

Nov. 28, 2023

Heidelberg Geometry Seminar

Heidelberg, Germany

The Steenrod problem for orbifolds and polyfold Gromov–Witten invariants

Dec. 10, 2020

Oberseminar Dynamische Systeme

Bochum, Germany

Gromov–Witten Axioms for Symplectic Manifolds via Polyfold Theory

Oct. 24, 2018

SISSA

Trieste, Italy

Naturality of Polyfold Invariants

May 2, 2018

Mathematical Sciences Research Institute

Berkeley, California

The Polyfold Gromov–Witten Invariants & Gromov–Witten Axioms for Symplectic Manifolds via Polyfold Theory

Apr. 25, 2018

Berkeley Topology Seminar

Berkeley, California

PROJECTS

LLMs from the Mathematical Viewpoint

www.wolfgangschmaltz.com/llm_mathematics

- A mathematics first presentation of the fundamental theory of Large Language Models, designed specifically for an audience of mathematicians. Using mathematical terminology and formula, explains the definitions and constructions of tokenizers, causally masked attention, the transformer architecture, and a language model itself. Basic implementations of these constructions in Pytorch are also included to act as a bridge between the abstract mathematical formulations and the implementation in code.

QLoRA Fine-Tuning to Generate Stories

www.wolfgangschmaltz.com/stories

- A QLoRA adapted small language model which generates stories based upon discrete user-specified criteria: theme, topic, word count, and complexity. Based on a small (35 million parameters) language model, and trained using supervised fine-tuning, QLoRA, and quantization. A user interface for interacting with the model, including a text streaming effect, was built using streamlit. Accessible as a webapp, hosted on huggingface, streamlit, and my personal website.

SKILLS

Programming Languages: Python (excellent), SQL (excellent), Java, Bash

Foreign Languages: English (native), German (C1), Turkish (A2)